

Lists and List Operations

The World Is Not Flat



The World Is Not Flat



	A	B	C	D	E
1	COUNTRY	CHOCOLATE	NOBEL	POPULATION	INTERNET
2	Australia	4.5	5.5	22	79.5
3	Austria	10.2	24.3	8	79.8
4	Belgium	4.4	8.6	11	78.0
5	Brazil	2.9	0.1	197	45.0
6	Canada	3.9	6.1	34	83.0
7	China	0.7	0.1	1344	38.3
8	Denmark	8.5	25.3	6	90.0
9	Finland	7.3	7.6	5	89.4
10	France	6.3	9.0	65	79.6
11	Germany	11.6	12.7	82	83.0
12	Greece	2.5	1.9	11	53.0
13	Ireland	8.8	12.7	5	76.8
14	Italy	3.7	3.3	61	56.8
15	Japan	1.8	1.5	128	79.1
16	Netherlands	4.5	11.4	17	92.3
17	Norway	9.4	25.5	5	94.0
18	Poland	3.6	3.1	39	64.9
19	Portugal	2.0	1.9	11	57.8
20	Spain	3.6	1.7	46	67.6
21	Sweden	6.4	31.9	9	94.0
22	Switzerland	11.9	31.5	8	85.2
23	UK	9.7	18.9	63	86.8
24	USA	5.3	10.8	312	77.9

Many of us started in the world of **flat files**.

The world of data is much more diverse.

The **backbone** of most data structures is the **list**.

The World Is Not Flat



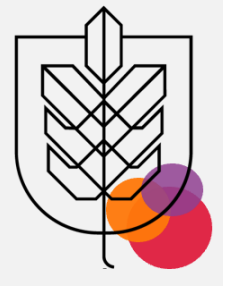
If this were a list,

	A	B	C	D	E
1	COUNTRY	CHOCOLATE	NOBEL	POPULATION	INTERNET
2	Australia	4.5	5.5	22	79.5
3	Austria	10.2	24.3	8	79.8
4	Belgium	4.4	8.6	11	78.0
5	Brazil	2.9	0.1	197	45.0
6	Canada	3.9	6.1	34	83.0
7	China	0.7	0.1	1344	38.3
8	Denmark	8.5	25.3	6	90.0
9	Finland	7.3	7.6	5	89.4
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20	Spain	3.6	1.7	46	67.6
21	Sweden	6.4	31.9	9	94.0
22	Switzerland	11.9	31.5	8	85.2
23	UK	9.7	18.9	63	86.8
24	USA	5.3	10.8	312	77.9

It would look like this:

```
[  
  ["Australia", 4.5, 5.5, 22, 79.5],  
  ["Austria", 10.2, 24.3, 8, 79.8],  
  ["Belgium", 4.4, 8.6, 11, 78.0],  
  ["Brazil", 2.9, 0.1, 197, 45.0]  
  ...  
]
```

The World Is Not Flat



```
[ ["Australia", 4.5, 5.5, 22, 79.5] , ["Austria", 10.2, 24.3, 8, 79.8] ,  
  ["Belgium", 4.4, 8.6, 11, 78.0] , ["Brazil", 2.9, 0.1, 197, 45.0] ... ]
```

Notice how **we have no headers**.

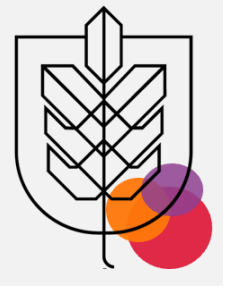
This is a **common situation** when scraping data from the internet.

Lists of Lists

The World Is Really Not Flat



Lists of Lists



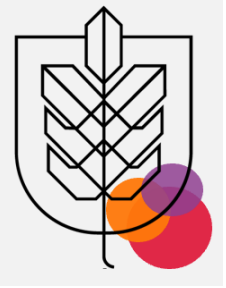
Oftentimes, we group things together in sub lists (lists of lists)

```
[ [“Australia”, 4.5, 5.5, 22, 79.5] , [“Austria”, 10.2, 24.3, 8, 79.8] ,  
  [“Belgium”, 4.4, 8.6, 11, 78.0] , [“Brazil”, 2.9, 0.1, 197, 45.0] ... ]
```

Advantages:

- helps keep data organized
- easy to store in databases

Accessing Elements in Lists of Lists



```
country_list = [ ["Australia", 4.5, 5.5, 22, 79.5] ,  
                  ["Austria", 10.2, 24.3, 8, 79.8] ,  
                  ["Belgium", 4.4, 8.6, 11, 78.0] ,  
                  ["Brazil", 2.9, 0.1, 197, 45.0] ... ]
```

country_list[0] = gives us our first sublist

country_list[0] = ["Australia", 4.5, 5.5, 22, 79.5]

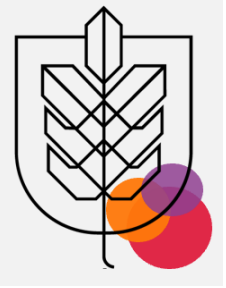
Accessing Elements in Lists of Lists



```
country_list = [ ["Australia", 4.5, 5.5, 22, 79.5] ,  
                  ["Austria", 10.2, 24.3, 8, 79.8] ,  
                  ["Belgium", 4.4, 8.6, 11, 78.0] ,  
                  ["Brazil", 2.9, 0.1, 197, 45.0] ... ]
```

```
country_list[3] = ["Brazil", 2.9, 0.1, 197, 45.0]
```


Accessing Elements in Lists of Lists

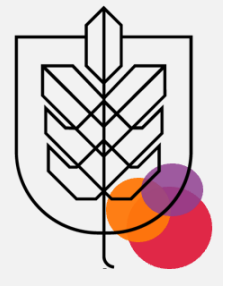


```
country_list = [ ["Australia", 4.5, 5.5, 22, 79.5] ,  
                  ["Austria", 10.2, 24.3, 8, 79.8] ,  
                  ["Belgium", 4.4, 8.6, 11, 78.0] ,  
                  ["Brazil", 2.9, 0.1, 197, 45.0] ... ]
```

`country_list[0][0]` = gives us the first element of our
first sublist

`country_list[0][0]` = "Australia"

Accessing Elements in Lists of Lists



```
country_list = [ ["Australia", 4.5, 5.5, 22, 79.5] ,  
                  ["Austria", 10.2, 24.3, 8, 79.8] ,  
                  ["Belgium", 4.4, 8.6, 11, 78.0] ,  
                  ["Brazil", 2.9, 0.1, 197, 45.0] ... ]
```

```
country_list[1][3] =      8
```

Accessing Elements in Lists of Lists



```
country_list = [ ["Australia", 4.5, 5.5, 22, 79.5] ,  
                 ["Austria", 10.2, 24.3, 8, 79.8] ,  
                 ["Belgium", 4.4, 8.6, 11, 78.0] ,  
                 ["Brazil", 2.9, 0.1, 197, 45.0] ... ]
```

`country_list[0:2]` = gives us our first two sublists

```
country_list[0:2] = ["Australia", 4.5, 5.5, 22, 79.5] ,  
                  ["Austria", 10.2, 24.3, 8, 79.8 ]
```



.....

```
country_list[1:3] = ["Austria", 10.2, 24.3, 8, 79.8] ,  
                    ["Belgium", 4.4, 8.6, 11, 78.0]
```

Python Time!

